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Area calculator module for geometric area calculations.

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Provides the [AreaCalculator](#) class with utility methods for computing the area of basic geometric shapes. This initial version supports rectangle area calculations based on a given length and width.

Typical usage would be:

```
from area_calculator import AreaCalculator
```

```
calc: AreaCalculator = AreaCalculator()
```

```
area: float = calc.calculate_rectangle_area(length=5.0, width=3.0)
```

```
print(area)
```

Classes

[builtins.object](#)

[AreaCalculator](#)

```
class AreaCalculator(builtins.object)
```

```
AreaCalculator() -> None
```

Provides utility methods for geometric area calculations.

An [AreaCalculator](#) instance exposes methods for computing the area of geometric shapes. The class itself holds no state; each method operates purely on the parameters provided to it. This mirrors the Java utility class pattern but in Python.

Attributes:

None. [AreaCalculator](#) is a stateless utility class.

Methods defined here:

```
__init__(self) -> None
```

Create a new [AreaCalculator](#) instance.

The constructor takes no arguments and performs no setup, as the class is stateless. It exists to mirror the Java [AreaCalculator](#)() default constructor.

```
calculate_rectangle_area(self, length: float, width: float) -> float
```

Calculate the area of a rectangle based on its length and width.

Multiplies the two supplied dimensions and returns the result. Both inputs are expected to be non negative floating point values representing the rectangle's sides in the same unit.

Parameters:

length : float

The vertical or horizontal extent of the rectangle.

width : float

The horizontal or vertical extent of the rectangle.

Returns:

float

The total area as the product of length and width.

Data descriptors defined here:

```
__dict__
```

dictionary for instance variables

```
__weakref__
```

list of weak references to the object

Functions

main() -> None

Run the interactive [AreaCalculator](#) menu loop.

Displays a menu that lets the user calculate a rectangle's area by entering a length and a width, or exit the program. The loop continues until the user selects the exit option.